

Abstract No.

Chromatin Deregulation in Glioblastoma and Applications for Liquid Biopsies

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My project investigates chromatin dynamics in glioblastoma and their application to cell-free DNA (cfDNA) based diagnostics using liquid biopsies. I examine differences in nucleosome positioning between glioblastoma patients and healthy individuals, focusing on dyad position, occupancy, and stability, and on their association with epigenetic modifications. This approach aims to improve understanding of genome organisation and identify disease-associated genomic regions.

Plasma-derived cfDNA provides a minimally invasive alternative to tissue biopsies, which are often risky in glioblastoma. It enables real-time monitoring of tumour-associated genetic and epigenetic changes, supporting assessment of disease progression and treatment response. Given the high recurrence rate of glioblastoma, cfDNA analysis is particularly valuable for detecting minimal residual disease and early relapse.

The project also investigates tumour heterogeneity by comparing cfDNA-derived signals with those obtained from tissue-based biopsies. Additionally, I develop machine learning models using cfDNA-derived features for glioblastoma classification, creating diagnosis frameworks with potential applications beyond this cancer type.

Overall, this research integrates genomic and computational methods to advance non-invasive cancer diagnostics and personalised medicine in glioblastoma.



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